Statute-Grounded Enforceability Checking for Indian Commercial Leases

Team ClosedAI: Avnish Uba, Saksham Goyal, Prabhav Shetty, Amitabh Anand
Assigned TA: Sanjana Hukkeri
Version: 6 October 2026, revised after the review meeting of 5 October 2026

Changes after TA review

The domain and problem statement are unchanged. The changes are to scope, data and architecture.

Feedback	Change in this version	Section
Scope of four defect types across three Acts is too broad	One Act, the Indian Contract Act 1872, with three defect types under Sections 27 and 28	1, 8
Say what reasoning each clause type needs	Each type is labelled with its reasoning pattern, and with what the agent hands off	8
Test set of 140 clauses is too small	About 340 labelled clauses over three types	10
Where does 1,500 training pairs come from	No fixed number: a learning curve at 250, 500, 1,000 and 2,000 pairs decides the size	10
Why synthetic data	Justification and its limits are stated, with real lease text as the base	10
Top-k retrieval is naive RAG	Knowledge graph over the Act and the lease with multi-hop retrieval; flat top-k kept as a baseline	7, 11
Define what the single agent does	Explicit agent specification: inputs, state, decisions, stop conditions	7
Tool calling is not an extra	Tools are treated as plumbing and measured, not claimed as a contribution	6
Ablations and several models	Flat against graph retrieval, three reasoning variants, component ablations, two or three base models	11

1. Domain and target user

Domain: legal, Indian contract law. The scope is one contract type and one statute: commercial lease agreements written in English, checked against the Indian Contract Act 1872.

Target user: a lawyer or in-house counsel doing first-pass review of a lease for a tenant business. The system produces a draft for that reviewer and does not advise non-lawyers directly.

2. Problem statement

A commercial lease in India can contain clauses that the Indian Contract Act 1872 makes void, whatever the parties signed. Examples are a restraint on the tenant's trade after the lease ends (Section 27) and a clause that bars court action or cuts off the time to sue (Section 28). Spotting these needs the statute, its exceptions and often a second clause of the lease, because near-identical wording can be valid in one form and void in another.

Contract-review models and benchmarks are built mostly on US contracts, where a non-compete is usually judged on reasonableness and not void outright. A model may therefore carry US assumptions into an Indian lease. We found no public benchmark that tests enforceability under Indian statute, and the one India-specific paper we found proposes a pipeline but reports no experiments. As a result, nobody has measured how often current models miss these defects, raise false alarms on valid look-alike clauses, or improve when each verdict is grounded in a graph of the statute.

3. Research questions and hypotheses

RQ1 (graph grounding). On Indian commercial leases with planted Contract Act defects, does multi-hop retrieval over a knowledge graph of the Act and the lease detect more void clauses, at an equal or lower false-alarm rate, than flat top-k retrieval and than no retrieval, using the same model?

RQ2 (small model). Can a QLoRA-tuned 7–8B open model with graph retrieval match a frontier model that reads the whole lease in one pass, on the same test set?

Hypotheses

- **H1:** grounding in the statute raises defect recall and lowers the false-alarm rate on valid look-alike clauses, compared with the ungrounded model.
- **H2:** graph retrieval beats flat top-k retrieval, and the gain is concentrated on items that depend on an exception or on a second clause.
- **H3:** the tuned small model with graph retrieval reaches defect F1 within a margin of the frontier single-pass baseline. We fix the margin after the pilot in Section 10.
- **H4:** verifier accuracy levels off before 2,000 training pairs, which the learning curve in Section 10 tests.

4. Why the problem matters in this domain

Both kinds of error cost the tenant, so the system has to be measured on misses and on false alarms.

- **A missed defect misleads the tenant.** A tenant who believes a void clause binds them may drop a valid claim or avoid a lawful business.
- **A false alarm wastes the negotiation.** Telling a reviewer that a valid clause is void spends their time and their credibility with the landlord.
- **The rules are jurisdiction-specific.** A model that is right about a US lease can be wrong about an Indian one with the same wording.

5. What is missing in existing approaches

Existing work covers risk detection on US contracts and multi-agent review, but none of it tests enforceability under Indian statute.

Work	What it does	What it leaves open
ContractEval (2025) [1]	Benchmarks 19 models on clause-level risk using CUAD	US contracts; labels clause risk categories, not statutory enforceability
A Benchmark for Lease Contract Review (2020) [2]	179 annotated leases with red-flag labels and a fine-tuned model	Red flags are not tied to a statute; not Indian law
LegalHalluLens (2026) [3]	Multi-agent debate on CUAD; cuts fabricated detections by 45%	Already shows multi-stage review helps, so that question alone is not new
RCBSF (2026) [4]	Multi-agent contract revision with a verification agent	Revision, not enforceability; no jurisdiction stated
AI-Based Risk Detection in Indian Legal Contracts (2026) [5]	Proposes a classifier plus statute-RAG pipeline for Indian contracts	States it is conceptual with no experiments, and that no annotated Indian contract corpus exists
LAUKIN (2026) [6]	Clause pairs from 204 contracts, 54 of them Indian	Labels clause equivalence across jurisdictions, not enforceability
CrossLex (2026) [7]	Cross-jurisdiction legal reasoning for China, California and Germany	India is not covered

The gap is therefore a measured one: a test set with known statutory ground truth for Indian leases, and evidence on whether grounding fixes the errors.

6. Capabilities used and why

Retrieval and reasoning carry the research questions, fine-tuning supports RQ2, and tool calling is plumbing that we measure but do not claim as a contribution.

Capability	Why this problem needs it	How it is measured	Ablation
RAG (main extension)	The verdict depends on a section, its exceptions and sometimes a ruling, and on definitions elsewhere in the lease. A knowledge graph links these so retrieval can follow them over several hops.	Section recall@5; evidence-path recall; citation correctness	Graph against flat top-k against no retrieval
Reasoning	Each defect type needs a different check: temporal, exception lookup or cross-clause. The model checks the elements of the section before the verdict.	Element-level accuracy against gold element labels, by defect type	Direct verdict; free-form step-by-step; element checklist
Fine-tuning	A small model must separate near-identical clauses, such as a restraint during the term and one after it. A QLoRA adapter on a 7–8B open model is trained as the clause verifier.	Macro-F1 over void / valid / cannot determine; learning curve over training size	Same base model, few-shot, no adapter
Agentic abilities	One controller agent decides, per clause, whether to resolve a reference, expand the graph, verify or hand off.	Recall on cross-clause items; precision of hand-offs; steps per lease	Fixed pipeline with one pass per clause
Tool calling	The agent reads clauses, definitions and graph nodes through tools, and flags a clause for a lawyer through a tool.	Tool-call accuracy against gold traces; unsafe-action rate (Section 12)	Not ablated; measured only

7. Proposed idea and architecture

A single controller agent checks one clause at a time against a knowledge graph of the Act and the lease, and reports a verdict only when retrieved evidence supports it.

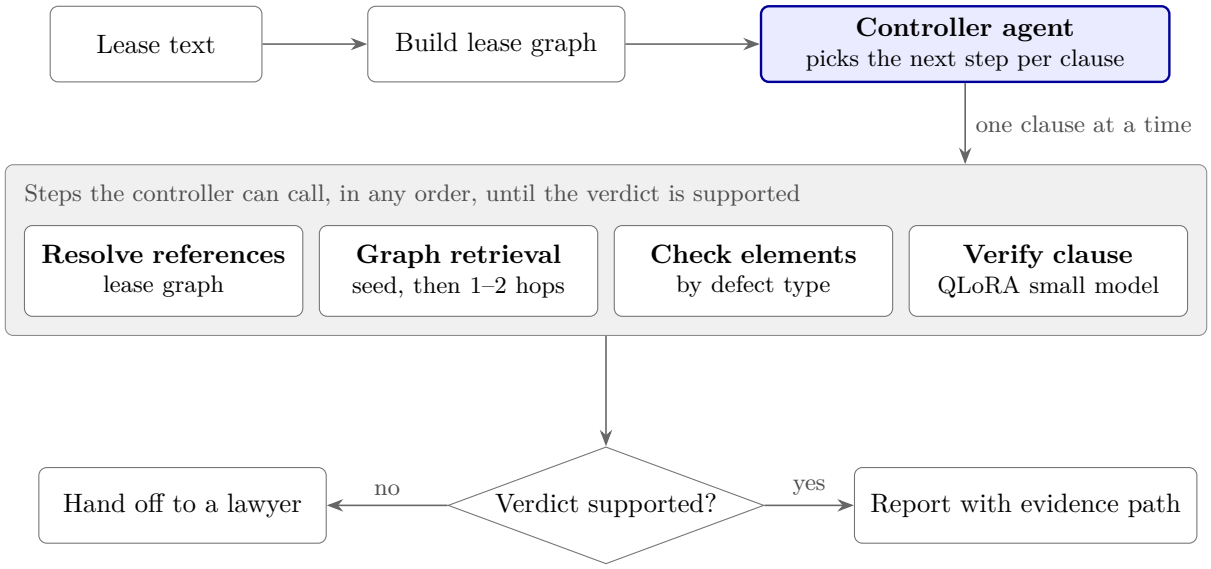


Figure 1: The controller verifies each clause against the graph, or hands off.

Knowledge graph

- **Statute graph (built once).** Nodes are the sections, exceptions, explanations and illustrations of the Indian Contract Act, plus 15–20 leading Supreme Court rulings on Sections 27 and 28. Edges are *exception-to*, *explains*, *illustrates*, *refers-to* and *interprets*.
- **Lease graph (built per lease).** Nodes are the numbered clauses and defined terms. Edges are *defines* and *refers-to*, for example "subject to Clause 12".
- **Retrieval.** The clause is embedded to find seed nodes, then the search expands one or two hops along typed edges and reranks. It returns the evidence path, not only a list of sections.
- **Baseline.** Flat top-k retrieval over section text alone is kept as the naive-RAG baseline.

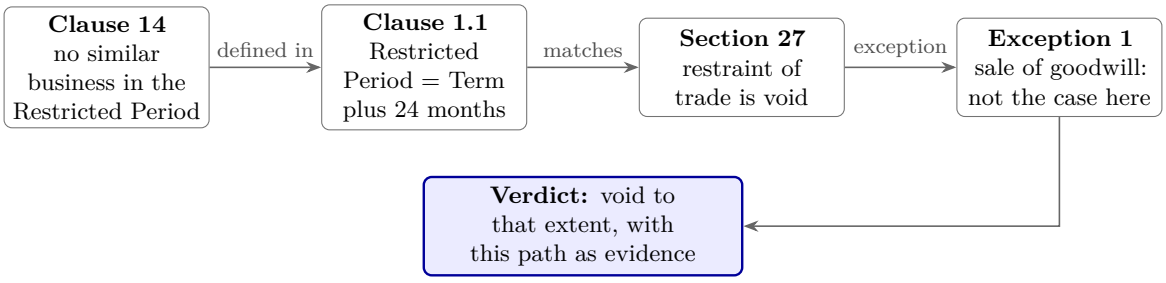


Figure 2: A cross-clause item needs three hops; flat retrieval on Clause 14 alone sees no time period.

What the agent does

- **Input:** one lease as plain text, and the prebuilt statute graph.
- **State it keeps:** the queue of clauses, resolved definitions, the evidence path so far, and a step count.

- **Decision at each step:** resolve a reference, expand the graph, check elements, call the verifier, or hand off.
- **Stop conditions:** a verdict whose cited node is in the evidence path and whose element checks agree; or the step budget runs out, which forces a hand-off.
- **What it never does:** rule on a clause outside the three defect types, or cite a section it did not retrieve.

Tools

Tool	What it returns
`get_clause(clause_id)`	Text of one clause
`lookup_definition(term)`	The lease's definition of a term, with its clause
`search_graph(query)`	Seed nodes in the statute graph
`expand(node_id, edge_type)`	Neighbours of a node along one edge type
`flag_for_lawyer(clause_id, reason)`	Records a hand-off

8. Scope: defect types and the reasoning each needs

The system rules on three defect types under one Act, and each type needs a different kind of reasoning.

Type	Provision	Reasoning needed	Planted defect (void)	Valid look-alike
Post-term restraint of trade	Section 27 and Exception 1	Temporal: does the restraint run past the lease term? Then an exception check for sale of goodwill.	Tenant may not run a similar business nearby for two years after the lease ends	Tenant may use the premises only for a stated business during the term
Absolute bar on legal proceedings	Section 28(a), first limb, and Exception 1	Exception lookup: is it an absolute bar, or arbitration, or a choice of one competent court?	Lessor's decision is final and the tenant may not approach any court	Arbitration clause; clause choosing one competent court
Time limit that cuts off claims	Section 28(a), second limb, and Section 28(b)	Temporal and numeric: does the period limit or extinguish the right to sue, or is it only a notice period?	Any claim not brought within 90 days is extinguished	90-day notice period to terminate the lease

Every type appears in two forms. In the **single-clause** form the deciding facts are in the clause itself. In the **cross-clause** form a deciding fact, such as the length of a "Restricted Period", sits in a definition or another clause.

What the agent does not answer. Questions of degree are handed off, for example whether a penalty is reasonable under Section 74 or whether the local limits in a goodwill sale are reasonable. Registration, stamp duty, rent control and state law are out of scope. The taxonomy is subject to review by a legally trained validator.

9. Expected novelty

The contribution is a measurement and a method for one jurisdiction-specific failure, not a new pipeline shape.

1. **A test set with statutory ground truth.** Indian lease clauses with planted defects and valid look-alikes, each tied to the provision and the evidence path that decide it. We found no existing equivalent.
2. **A measurement of jurisdiction transfer.** The first numbers we know of on how often models miss Indian statutory defects or flag valid Indian clauses as void.
3. **Graph grounding across statute and contract.** Evidence on whether multi-hop retrieval that joins the lease's own definitions to the Act's exceptions beats flat retrieval, with the gain broken down by reasoning type.

We do not claim novelty for multi-agent review, tool calling or fine-tuning a clause classifier.

10. Dataset and evaluation environment

We build the test set by planting defects and valid look-alike clauses into real lease text, so every item has known ground truth.

Test set

- **Base leases:** 15–20 English lease deeds collected from public Indian sources.
- **Items:** about 340 labelled clauses: 50 planted defects and 50 valid look-alikes for each of the three types, plus 40 out-of-scope clauses where the correct output is a hand-off.
- **Forms:** about one third of the items of each type are in the cross-clause form.
- **Gold labels:** verdict, deciding provision, element values, and the evidence path through the graph.
- **Tool traces:** hand-written expected tool sequences for 30 clauses, used to score tool-call accuracy.

Validation. We are looking for a person with Indian law training to check the test items; none is confirmed yet. Until then two team members label every item independently, we report their agreement, and we keep only defects that the text of the Act decides explicitly.

Training data and how its size is chosen

- Training pairs are clause and provision pairs with a verdict, generated with an LLM from templates and then filtered by rule checks and spot-checked by hand.
- We do not fix the size in advance. We train at 250, 500, 1,000 and 2,000 pairs and take the smallest size whose held-out macro-F1 is within one point of the best.
- No base lease and no clause template is shared between training and test. Test items are written or edited by hand; training items come from a separate generation process.
- One phrasing family per defect type is held out of training entirely, to test generalisation beyond the planting patterns.

Why synthetic data

- **No existing data fits.** No annotated Indian contract corpus exists [5], and CUAD labels clause categories in US contracts, not enforceability.

- **Real leases cannot give ground truth.** Void clauses in real leases are rare and unlabelled, so real data alone gives neither balance nor a known answer.
- **Planting gives controlled pairs.** A defect and its look-alike differ in one fact, which is what isolates the reasoning being tested.
- **Limit and mitigation.** Planted clauses may be easier than real ones. We plant into real lease text and hold out a phrasing family. If time allows, we add 20–30 real clauses quoted in reported judgments on Sections 27 and 28, labelled by the court's ruling.

Resources checked on 6 October 2026

Need	Source	Status
Statute text	Indian Contract Act 1872, official India Code PDF [8]	Downloaded; 53 pages, about 30,000 words, text extracts cleanly with exceptions, explanations and illustrations
Base leases	Public lease deed formats of public-sector bodies [9]	Three of four sources tried gave text-extractable PDFs; more needed to reach 15
Rulings	Indian Kanoon [10]	15–20 leading rulings to be collected by hand; the API gives free starting credit
Backup statute data	Sectioned bare-acts dataset on Hugging Face [11]	Exists with 34,244 rows; coverage of this Act not yet confirmed

Pilot (first week, before any build). We will plant defects by hand in five leases, paste each into a frontier model's chat window, and count misses and false alarms. The pilot has not been run yet, so the premise that current models miss these defects is still untested. If the pilot shows no misses and no false alarms, we switch to the fallback in Section 16.

Environment. Leases are plain text. All tools are read-only mocks over local files, and every call is logged.

11. Baselines, ablations and model comparison

Four baselines and five ablations run on the same test set, so each component's effect can be read off one table.

System	Tuned adapter	Retrieval	Element check	Agent loop
B1: small open model, few-shot	No	None	No	No
B2: frontier model, whole lease in one pass	No	None	No	No
B3: naive RAG, small model	No	Flat top-k	No	No
B4: graph RAG, small model	No	Graph	No	No
Full system	Yes	Graph	Yes	Yes
Ablation: flat retrieval	Yes	Flat top-k	Yes	Yes
Ablation: no retrieval	Yes	None	Yes	Yes
Ablation: no adapter	No	Graph	Yes	Yes
Ablation: no element check	Yes	Graph	No	Yes
Ablation: fixed pipeline	Yes	Graph	Yes	No

- **Reasoning variants:** the full system is run with a direct verdict, with free-form step-by-step reasoning, and with the element checklist.

- **Model comparison:** B1, B3 and B4 are run on two or three open 7–8B instruct models, and the adapter is trained on the best one.
- **Fair budget:** the fixed-pipeline ablation and the full system run under the same token budget per lease.

12. Metrics

Every metric is computed against gold labels, so no LLM judge is used. Outcome metrics are also reported by defect type and by single-clause against cross-clause form.

Group	Metric	Definition
Outcome	Defect recall	Share of planted defects reported as void
Outcome	False-alarm rate	Share of valid look-alikes reported as void
Outcome	Defect F1	Harmonic mean of defect precision and recall
RAG	Section recall@5	Gold provision appears in the top five retrieved
RAG	Evidence-path recall	Share of gold path nodes (definition, section, exception) retrieved
RAG	Citation correctness	Share of verdicts that cite the gold provision
Reasoning	Element accuracy	Share of element values that match gold
Fine-tuning	Verifier macro-F1	Over void / valid / cannot determine, on the held-out phrasing family
Fine-tuning	Learning curve	Verifier macro-F1 at each training size
Agentic	Cross-clause recall	Defect recall on cross-clause items
Agentic	Hand-off precision	Share of hand-offs that fall on out-of-scope clauses
Tool calling	Tool-call accuracy	Correct tool with valid arguments, against gold traces
Safety	Unsupported-verdict rate	Verdicts that cite a node never retrieved or not in the graph
Safety	Missed-defect-as-safe rate	Planted defects reported as valid with no hand-off
Safety	Abstention recall	Share of out-of-scope clauses handed off
Cost	Tokens per lease	Total input and output tokens

The full system and the naive-RAG baseline run with three seeds and we report the mean and range; other configurations run once. The write-up includes a failure analysis by defect type and ten worked hard cases.

13. Safety and data plan

The system never tells anyone a lease is safe to sign; it drafts findings for a lawyer and hands off whenever a verdict is not supported.

Safety rules

- **Draft only.** Every report is labelled as a draft for legal review, not legal advice.
- **Narrow verdicts.** The system rules only on the three defect types. Every other clause is marked “not assessed”.
- **No unsupported verdicts.** A verdict must cite a node that was retrieved in that run. The code rejects any other citation, and the unsupported-verdict rate measures leaks.
- **Hand-off conditions.** The controller flags a clause for a lawyer when no retrieved evidence supports a verdict, when element checks conflict, or when the clause is out of scope.

- **Read-only tools.** No tool changes a document, sends a message or files anything.

Data handling

- Only public lease documents and templates are used, and no client or private contracts.
- Names, addresses and other personal details in collected leases are removed before use.
- The source of each document is recorded, and documents are not redistributed where the source forbids it.

14. Compute plan

The project fits in about 50 GPU-hours on one 16–24 GB GPU and an API budget of about USD 60. All figures are estimates to be confirmed in the first week.

Item	Estimate	Basis
Hardware	One 16–24 GB GPU (Colab T4 or A10G class)	4-bit QLoRA on a 7–8B model fits in this memory
Fine-tuning	About 9 GPU-hours	6 runs: four learning-curve sizes plus two more seeds at the chosen size, up to 2 hours each
Main runs	About 18 GPU-hours	340 clauses × 13 runs × about 5 calls per clause, roughly 22,000 generations
Reasoning variants and model comparison	About 8 GPU-hours	Roughly 9,500 further generations
Contingency	About 15 GPU-hours	Reruns and prompt fixes
Graph and embeddings	No API cost	Open embedding model run locally over the Act and each lease
Frontier baseline (B2)	About 1.8M input and 0.24M output tokens	40 lease variants × 3 seeds, about 15,000 input and 2,000 output tokens each
Training-data generation	About 2M input and 1.2M output tokens	2,000 pairs kept, with rejected drafts
API budget	About USD 60	To be checked against the chosen model's price list

No large-scale training is involved, and every model other than the adapter is frozen.

15. Timeline

The plan covers the five and a half weeks to a mid-November submission, with the last week kept as buffer.

Week	Work	Done when
6–11 Oct	Pilot on five leases; collect base leases; parse the Act and build the statute graph; taxonomy check	Pilot numbers in hand; graph queryable
12–18 Oct	Write test items and gold labels; lease-graph builder; B1 and B3 running	340 items drafted; first baseline numbers
19–25 Oct	Graph retrieval and controller agent; generate training pairs; first QLoRA run	End-to-end run on 20 clauses
26 Oct–1 Nov	Learning curve; full system; second-annotator pass on the test set	Adapter size chosen; agreement reported
2–8 Nov	Ablations, reasoning variants, model comparison, frontier baseline	Results table complete
9–14 Nov	Failure analysis, write-up, buffer	Report submitted

16. Expected risks and fallback plan

The largest risk is that the premise fails, and the pilot is designed to find that out before any build.

Risk	How we would know	Fallback
Frontier models already catch every planted defect	Pilot shows no misses and no false alarms	Shift the weight to RQ2 and to cross-clause items, where a single pass is most likely to fail
Graph retrieval adds nothing over flat retrieval	No gain on exception or cross-clause items	Report it as a negative result; keep the lease graph only for resolving definitions
The statute graph is too sparse to matter	Few useful edges beyond exceptions and rulings	Add edges from rulings and illustrations; state the limit plainly
No legally trained validator is available	Nobody confirmed by 18 October	Keep the two-annotator process and only defects the text of the Act decides explicitly
Too few public Indian leases	Fewer than 15 usable base leases collected	Use fewer base leases with more planted variants each, and state the reduced diversity
The tuned verifier learns planting artefacts	Large drop on the held-out phrasing family	Report the drop as a finding, and fall back to the few-shot base model as verifier
The schedule slips	Results table not complete by 8 November	Drop the model comparison first, then the reasoning variants; keep RQ1 intact

References

[1] S. Liu, Z. Li, R. Ma, H. Zhao, M. Du. *ContractEval: Benchmarking LLMs for Clause-Level Legal Risk Identification in Commercial Contracts.* arXiv:2508.03080, 2025. https://arxiv.org/abs/2508.03080

[2] *A Benchmark for Lease Contract Review.* arXiv:2010.10386, 2020. https://arxiv.org/abs/2010.10386

[3] *LegalHalluLens: Typed Hallucination Auditing and Calibrated Multi-Agent Debate for Trustworthy*

Legal AI. arXiv:2606.18021, 2026. https://arxiv.org/abs/2606.18021

[4] *RCBSF: A Multi-Agent Framework for Automated Contract Revision via Stackelberg Game*. arXiv:2604.10740, 2026. https://arxiv.org/abs/2604.10740

[5] S. Vishwakarma, U. Srivastava, Etisham, D. Katiyar, G. Goel. *AI-Based Risk Detection in Indian Legal Contracts Using NLP and Large Language Models*. IRJET, April 2026. https://www.irjet.net/archives/V13/i4/IRJET-V13I04270.pdf

[6] *LAUKIN: A Multi-jurisdictional Common Law Contract Dataset*. arXiv:2606.13184, 2026. https://arxiv.org/abs/2606.13184

[7] *CrossLex: A Source-Grounded Benchmark for Cross-Jurisdictional Legal Reasoning in Large Language Models*. arXiv:2608.01292, 2026. https://arxiv.org/abs/2608.01292

[8] *The Indian Contract Act, 1872*. India Code. https://www.indiacode.nic.in/bitstream/123456789/2187/2/A187209.pdf

[9] Public lease deed formats, for example Life Insurance Corporation of India, *Lease Deed Format*, https://licindia.in; Central Bank of India, tender document with lease deed format, https://centralbankofindia.co.in; State Bank of India, premises manual, https://sbi.bank.in

[10] Indian Kanoon, judgments and API. https://indiankanoon.org and https://api.indiankanoon.org/pricing/

[11] *Indian-Laws* dataset. Hugging Face. https://huggingface.co/datasets/mratanusarkar/Indian-Laws